BANSEEDHAR GONDALIYA

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EXPERIENCE

IDFC FIRST Bank
April 2026 - current

ML Engineer | Gen AI Team — Bengaluru, India

- Building enterprise-grade **Agentic AI** and **GenAI** solutions using LangGraph, DeepAgents, AWS AgentCore, and LLM orchestration frameworks to enable intelligent automation across banking applications.

Samsung Research and Development
August 2023 - March 2026

Senior Machine Learning Engineer - Full-time | AI Productization Team — Bengaluru, India

- **Generative AI & Agentic AI:**
  - Engineered a production-scale Screen **AI Agent** using hybrid **RAG** (vector + keyword retrieval) and **LLM**-based task planning to interpret UI context and deliver dynamic interactive guidance, reducing task completion time by **55%** and enabling commercial deployment alongside an **ICVGIP 2025** publication and a granted Samsung **A1 patent**.
  - Fine-tuned the **Qwen2.5-Coder-32B-Instruct** using full fine-tuning and **LoRA** to enable structured UI reasoning and generation, empowering **agentic systems** to convert natural language plans into dynamic, interactive **UI workflows**.
- **Natural Language Processing(NLP):**
  - Designed and implemented an **entity classification model** for user interaction commands (address, email, event, phone, URL, etc.) using **ReTVec, TinyBERT, and LSTM**, achieving **96%** accuracy; commercialized in the ODCR module with **9 multilingual support**.
  - Developed a **relationship prediction model** using conversational and behavioral data (calls, messages, interactions), leveraging **ML and NLP models (LSTM, BiLSTM, BERT)** to achieve **95%** accuracy across **7 classes**; the innovation resulted in a **granted patent**.
- **Computer Vision (CV):**
  - Contributed to the development of **Screen AI and AI Select** technologies using on-screen **object detection** (e.g., timestamps, keyboards, maps) with **YOLO**, achieving an inference time of **150ms** and reducing the model size from **20MB to 1.5MB**.
  - Integrated the optimized UI element detection model into **On-Device Context Retriever (ODCR)** module to reduce textual noise and enhance **entity extraction** like phone numbers, emails, addresses, URLs, etc., enabling smart suggestions.

EDUCATION

National Institute Of Technology Surat
August 2019 - May 2023

*Bachelor of Technology in Computer Science and Engineering, CGPA: **9.04*** — Surat, India

PUBLICATIONS & ACHIEVEMENTS

Publication: "Screen2UI: Pixel-Only On-Device UI Hierarchy Generation," accepted at ICVGIP 2025.

Publication: "Who's Related? Fast and Accurate Family Relationship Detection in Conversations," accepted at ICASSP 2026.

Patent: "System and Method for Adaptive User Interface Analysis and Dynamic Interactive Task Guidance."

Samsung AI Expert (Workera): Core 1–4 certified (Mathematics for AI, ML, DL, CV, Diffusion Models, PyTorch, Prompt Engineering).

TECHNICAL SKILLS

Machine Learning, Deep Learning & AI: Agentic AI (LangGraph, LangChai, CrewAI, OpenAI-API, LlamaIndex, Retrieval-Augmented Generation (RAG), Pinecone, Vector DBs), NLP and Computer vision (Neural Networks, CNN, LSTM, Transformers, BERT), ML Supervised Learning, Large Language Models (LLMs), Generative AI, Named Entity Recognition (NER)

Programming Languages and Frameworks/Libraries: Python, C, C++, SQL, TensorFlow, PyTorch, Keras, Hugging Face, Scikit-learn, OpenCV, Agentic-AI (CrewAI, N8N), NumPy, Pandas, Matplotlib, Seaborn, NLTK, SpaCy

Big Data & Cloud Technologies: Google Cloud (Vertex AI), AWS (S3, SageMaker) , Hadoop, Spark

Model Deployment & Optimization: On-Device AI, TensorRT, TFLite, ONNX, Flask, FastAPI, Docker, Kubernetes

Additional Developer Tools: Git, Github, Linux, MySQL, MongoDB

PROJECTS

FinPilot Financial Advisor: Personalized Wealth Management through Multi-Agent AI

Multi Agent AI | LangGraph, LangChain, Pinecone, RAG, OpenAI API, LlamaIndex — **Github**

- Built a **multi-agent** Financial Advisor AI system with specialized **agents (Portfolio Analyzer, Finance Query Box, Watchlist Analyzer)** for intelligent investment insights, real-time monitoring, and personalized portfolio analysis.
- Implemented a scalable multi-agent orchestration framework using **LangGraph, OpenAI API, RAG, Binance API, Tavily web search, and financial document parsing,** reducing manual financial data processing effort by 60%.

POSITIONS OF RESPONSIBILITY

Mentor and Relationship Manager
May 2024 - Nov 2024

Samsung Research and Development | OnDevice AI Team

Mentorship of **2-Month** and **6-Month interns**, guiding them on AI/ML projects, coding best practices, and **industry workflows.**